



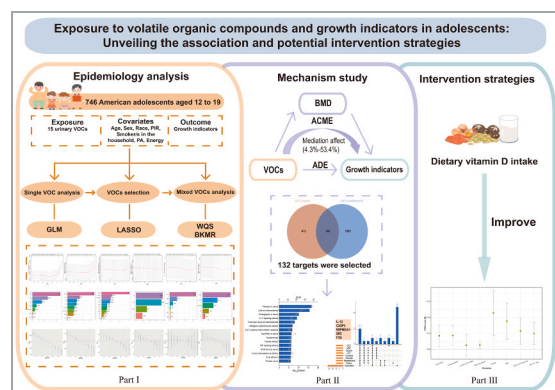
Exposure to volatile organic compounds and growth indicators in adolescents: Unveiling the association and potential intervention strategies

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GRAPHICAL ABSTRACT



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ABSTRACT

Environmental pollutant is considered to be one of the important factors affecting adolescent growth. However, the effects of volatile organic compounds (VOCs) exposure on adolescent growth have not been assessed. Data from the National Health and Nutrition Examination Survey (NHANES) 2011–2018 was used to examine the associations between VOCs exposure and adolescent growth indicators through three statistical models. The mediating effect of bone mineral density (BMD) on these associations was examined. The potential pathways and key targets were identified by the network pharmacology analysis methods. This study included 746 adolescents. Three statistical methods consistently showed a negative correlation between VOCs exposure and adolescent growth indicators. Furthermore, BMD mediated the relationship between VOCs exposure and adolescent growth indicators, with mediated proportion ranging from 4.3 % to 53.4 %. Network pharmacology analysis found a significant enrichment in IL-17 signaling pathway. Moreover, the adverse effects of VOCs exposure on adolescent growth were observed to significantly attenuate in adolescents with high serum vitamin D levels. Our results suggested that VOCs exposure was an adverse factor affecting adolescent growth, with BMD playing a significant

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regulatory role, and IL-17 signaling pathway was the underlying mechanism. Vitamin D supplementation may be a viable strategy to prevent VOCs exposure from affecting adolescent growth.

1. Introduction

Volatile organic compounds (VOCs) are a type of harmful low molecular mass carbon-containing compounds that are widespread in the environment. With the development of industrialization and urbanization, the sources of indoor and outdoor VOCs are gradually increasing. The major sources of personal exposure to VOCs include automotive exhaust, cooking fumes, cigarettes, insecticides, personal care products, and other industrial products [1–3]. According to reports, the human can absorb VOCs through many ways, such as air, food and skin [4,5]. Previous studies have shown that exposure to VOCs is closely related to the occurrence and development of several diseases, including cardiovascular and cerebrovascular diseases, cancer, hypertension, and respiratory diseases [6–9]. This emphasizes that the potential threat of VOCs exposure to human health cannot be ignored. Given that the biomarkers of VOCs exposure in urine have a longer biological half-life than that in blood and are more stable during transportation, storage and handling processes [10,11]. Therefore, the measurements of VOCs in urine have become reliable indicators for monitoring human VOCs exposure.

Adolescence is an important period of growth and development. The growth and development in adolescence are related to health throughout the life course [12]. Short stature and stunting have significant effects on morbidity and mortality in adulthood [13]. Many studies have shown that adolescent growth and development can be affected by diet, disease and environmental pollutants [14,15]. Environmental pollutants can produce multiple adverse effects in both humans and wildlife by interfering with the endocrine system of organisms [16,17]. Compared with adults, adolescents are more vulnerable to be affected by the adverse effects of environmental pollutants due to their higher metabolic and ventilation rates, and underdeveloped immune systems [18]. In a study of adolescents in the US, Wan et al. [19] found that exposure to higher PFAS levels affected growth and development in adolescents. Another cohort study of 212 Canadian adolescents showed that peri-pubertal period was a sensitive window for the effects of PCBs on growth [20]. Therefore, early detection of adverse factors affecting the growth and development of adolescents and effective interventions is crucial. Several studies have indicated that exposure to VOCs can affect lung growth and development, leading to compromised respiratory health in infancy [21,22]. However, there is still a lack of evidence regarding the effects of VOCs for adolescent growth.

Studying the mechanism of VOCs exposure on adolescent growth and development is crucial for developing effective prevention strategies. It is well-known that the accumulation of bone mineral density (BMD) is essential for growth and development. Adolescence is an essential period of bone development and maturity [23], and BMD continues to increase during this stage, reaching about 90 % of peak bone mass in late adolescence [24]. Existing studies have shown that there is a direct correlation between BMD and growth indicators such as height and weight [25,26]. Moreover, VOCs can affect BMD levels by affecting the activity of bone marrow cells, osteoblasts, and osteoclasts [27,28]. Under physiological conditions, osteoblasts, osteoclasts, and osteocytes collaborate to regulate the balance between bone resorption and bone formation, thereby facilitating the maintenance of regular bone development [29]. According to the study of Zhou et al. [30], among the VOCs that were related to BMD, N-acetyl-S-(N-methylcarbamoyl)-L-cysteine (AMCC), Mandelic acid (MA), N-acetyl-S-(2-cyanoethyl)-L-cysteine (CYMA), N-acetyl-S-(3-hydroxypropyl-1-methyl)-L-cysteine (HPMMA) and N-acetyl-S-(3,4-dihydroxybutyl)-L-cysteine (DHBMA) were the main contributors to the decrease of BMD. Based on the above, BMD reduction may be one biological mechanism of VOCs exposure on growth indicators of adolescents. Therefore, we

speculate that BMD, such as total BMD, subtotal BMD and lumbar spine BMD, may play a mediating role in the association between VOCs and growth indicators. To our knowledge, no studies have evaluated the mediation effects of BMD on the relationship between VOCs and growth indicators.

Vitamin D, a fat-soluble micronutrient, has various and important health benefits [31,32]. Some studies have shown that vitamin D deficiency can impair bone homeostasis [33,34]. And short stature in children is associated with lower levels of vitamin D [35]. An epidemiological study showed that higher serum 25 (OH) D concentrations were beneficial for increasing height and reducing the risk of low BMD in children aged 6–17 years [36]. However, some studies have yielded the exact opposite results. A meta-analysis of randomized controlled trials showed that vitamin D supplementation had little to no effect on linear growth in children [37]. Trilok-Kumar et al. [38] found that early vitamin D supplementation had no long-term effects on body composition and bone health. And Chowdhury et al. [39] also indicated that vitamin D status in early childhood was not associated with linear growth. Thus, the role of vitamin D in preventing the adverse effects of VOCs exposure on adolescent growth deserves further investigation.

In this study, we used a combination of epidemiological and functional enrichment analysis to explore the impact of VOCs exposure on adolescent growth indicators and its potential mechanisms, and proposing possible intervention strategies based on the mechanisms. Given that the measures taken to contain coronavirus disease 2019 may affect the correlation between VOCs exposure and growth indicators [40,41]. Also, the latest cycle of National Health and Nutrition Examination Survey (NHANES) data lacked BMD and serum vitamin D information. Therefore, we ultimately selected NHANES 2011–2018 for analysis. First, we used weighted generalized linear models (GLM), weighted quantile sum (WQS) and Bayesian kernel machine regression (BKMR) models to explore the relationship between VOCs exposure and adolescent growth indicators. GLM is a common regression analysis method used to analyze the association between a single component and health outcomes. WQS and BKMR models are two popular methods for assessing the combined effects of mixtures. Secondly, causal mediation analysis and network pharmacological analysis were used to explore the potential mechanism between VOCs exposure and adolescent growth indicators. Finally, we attempted to explore whether dietary intake of vitamin D can effectively prevent the adverse effects of VOCs exposure on adolescent growth indicators.

2. Methods

2.1. Study population

The NHANES is a cross-sectional survey conducted by the National Center for Health Statistics (NCHS) that assesses the health and nutritional status of the population in the US [42]. This study was conducted according to the guidelines laid down in the Declaration of Helsinki and all content involved was reviewed and approved by the NCHS Research Ethics Review Board. And participants provided written informed consent to participate in the study. Samples were extracted through multi-stage stratified sampling, which collects general demographic information, diet, laboratory examination and questionnaire data of the study population. This study used public data from NHANES 2011–2018. Of the 39,156 participants, 5215 were 12–19 years of age. Then, participants with incomplete information on growth indicators, urinary VOCs, urinary creatinine, BMD, serum vitamin D, and covariates (i.e. physical activity, poverty income ratio (PIR), and energy) were excluded. Ultimately, we included 746 adolescents in the analysis

(Fig. S1).

2.2. Urinary VOC metabolites measurement

The collected urine samples were stored at -20°C until they were shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta GA for analysis. As described by Alwis et al. [43], in the NHANES, urinary VOC metabolites were quantified with ultra performance liquid chromatography coupled with electrospray ionization tandem mass spectrometry (UPLC-EIS/MSMS). Chromatographic separation was performed by an Acquity UPLC HSS T3 column. For urinary VOC metabolites below the limits of detection (LOD), the concentration was expressed as $\text{LOD}/\sqrt{2}$. Detailed methods and information on urine VOC measurement can be found on the NHANES official website (<http://www.cdc.gov/nchs/nhanes/index.htm>).

The detection rate either too low or too high will affect the accuracy and stability of the results. We excluded urinary VOC metabolites with a detection rate of $< 80\%$ in consistence with previously published studies [44,45]. In our study, 15 metabolites were finally selected for analysis, including 2-methylhippuric acid (2MHA), 3-methylhippuric acid & 4-methylhippuric acid (3MHA+4MHA), N-acetyl-S-(2-carbamoyl-ethyl)-L-cysteine (AAMA), AMCC, 2-aminothiazoline-4-carboxylic acid (ATCA), N-acetyl-S-(benzyl)-L-cysteine (BMA), N-acetyl-S-(2-carboxyethyl)-L-cysteine (CEMA), CYMA, DHBMA, N-acetyl-S-(2-hydroxypropyl)-L-cysteine (2HMPA), N-acetyl-S-(3-hydroxypropyl)-L-cysteine (3HMPA), HPMMA, MA, N-acetyl-S-(4-hydroxy-2-butenyl)-L-cysteine (MHBMA3), and Phenylglyoxylic acid (PGA). To reduce errors in evaluating urinary VOC metabolites levels caused by external factors such as metabolic levels and water intake, effective urinary VOC metabolite concentrations were calibrated by urinary creatinine levels and expressed in $\mu\text{g/g}$ Cr. Abbreviations, parent compounds, LODs, detection rates and concentrations of each substance were shown in Table S1.

2.3. Body measurements

Body measurements, such as standing height, waist circumference, and weight, were obtained by health technicians at the Mobile Examination Center for each participant. Weight was measured using a digital scale and readings were recorded in kg. All health technicians were professionally trained. The equipment calibration was performed by health technicians. Detailed body measurement protocols, equipment, and quality control instructions can be found in the NHANES anthropometry procedures manual (https://www.cdc.gov/nchs/data/nhanes/2017-2018/manuals/2017_Anthropometry_Procedures_Manual.pdf). Body mass index (BMI) was calculated as weight in kilograms divided by the square of height in meters (kg/m^2). Height-for-age and BMI-for-age are identified as common growth indicators for adolescents by the World Health Organization. For this purpose, we also calculated the height-for-age and BMI-for-age of each participant using Centers for Disease Control and Prevention growth charts for ages 12–19 years [46].

2.4. BMD measurements

Dual-energy X-ray absorptiometry (DXA) is widely used for BMD measurement because of its simplicity, speed and minimal radiation exposure [47]. All eligible participants completed this measurement by experienced, trained and certified professional radiology technologists, while data of total BMD, subtotal BMD and lumbar spine BMD were collected. The detailed DXA measurements of each part to obtain information is available at <http://www.cdc.gov/nchs/nhanes/>.

2.5. Vitamin D intake assessments

The NHANES database obtained the dietary intake of each participant through a 24-hour dietary recall. Energy and nutrient composition

values were calculated using the US Department of Agriculture Dietary Sources of Nutrients database [48]. In this study, total daily intake of vitamin D was obtained by combining the first 24-hour dietary recall interviews to the daily intake of dietary supplements.

2.6. Serum vitamin D measurements

Briefly, the concentration of 25(OH)D in serum was detected by liquid chromatography-tandem mass spectrometry (LC-MS/MS). In this study, the total serum 25(OH)D was the result of adding the serum concentrations of 25(OH)D2 and 25(OH)D3. Based on the total serum 25(OH)D concentrations, we defined participants with serum vitamin D content less than 50 nmol/L as vitamin D deficiency group, and participants with serum vitamin D content not less than 50 nmol/L as vitamin D sufficiency group [49].

2.7. Covariates

Based on previous literature and the results of directed acyclic graph (DAG) analysis [50,51], we adjusted several variables as covariates to consider the potential confounders in our study: age, sex, race (Mexican American, other Hispanic, Non-Hispanic White, Non-Hispanic Black and other), smokers in the household, energy, physical activity (inactive or active), and PIR (Fig. S2). A household member who smokes more than or equal to one person was considered exposed to second-hand smoke. Based on the duration of moderate to vigorous physical activity per week, participants were classified as inactive (<10 min/week) and active (>10 min/week) [52].

2.8. Network pharmacological analysis

The candidate targets of VOCs were predicted using Similarity Ensemble Approach (SEA, <http://sea.bkslab.org/>) and PharmMapper (<http://lilab-ecust.cn/pharmmapper/index.html>). In the GeneCards database (<https://www.genecards.org/>), the protein targets related to adolescent growth and development were searched by setting the keyword “growth and development in adolescents (GaD in adolescents)”. Subsequently, Venn diagrams were plotted by R package “VennDiagram” and the intersection targets were inputted into the STRING database (<https://string-db.org/>) for constructing the protein-protein interactions (PPI). Then, the intersection targets were further ranked and screened by using the nine topological measures maximal clique centrality (MCC), maximum neighborhood component (MNC), degree, edge percolated component (EPC), closeness, radiality, betweenness, stress, and BottleNeck through the Analyze Network tool in the Cytoscape (Version 3.9.1). Kyoto Encyclopedia of Genes and Genomes (KEGG) and Gene Ontology (GO) analysis were performed by DAVID (<http://david.abcc.ncifcrf.gov/>). Significant GO terms and KEGG pathways were selected with P -value < 0.05 . The methodology was detailed in Fig. S3.

2.9. Statistical analysis

Descriptive statistics were performed on baseline features, where continuous variables were expressed as either mean (standard deviation, SD) or median (Q1, Q3), and categorical variables were represented by frequency (percentage, %). We performed natural logarithm (ln) transformations for 4 growth indicators (standing height, body weight, BMI, waist circumference) and urinary VOCs to better fit the normal distributions. Pearson's correlation was used to examine the correlation between multiple VOCs. For weighted analysis, we utilized the provided subsample weights from the VOCs data to better represent the true population of the US.

Firstly, we conducted weighted GLM to investigate the relationship between a single VOC exposure and growth indicators (standing height, weight, BMI, waist circumference, height-for-age and BMI-for-age). To

minimize potential collinearity of variables, the least absolute shrinkage and selection operator (LASSO) regression model was utilized. LASSO is a feature selection method that supports variable selection by penalizing the absolute values of the regression coefficients [44]. Through 10-fold cross-validation to select the penalty parameter λ , 1000 iterations were run to screen for key VOCs associated with growth indicators, aiming to minimize mean squared error. After screening several key VOCs associated with growth indicators, WQS regression and BKMR models were used to evaluate the impact of mixed exposure to VOCs on growth indicators. The models were adjusted for confounding factors as mentioned above. In the WQS model, the data was divided into training and testing sets in a 4:6 ratio and the model was established after bootstrapping with 2000 iterations. When the WQS index was significant, the respective weights were examined to determine the relative contributions of each VOC to the growth indicators within the index. However, WQS regression cannot evaluate the nonlinearity and interactions of mixtures [53,54]. Therefore, BKMR model was used to further analyze the mixed effects of VOCs on growth indicators. Specifically, the mixed effects of VOCs on growth indicators by comparing growth indicator estimates per 10th percentile increase/decrease from the median concentration (reference) of VOCs. The impact of each VOC on various growth indicators was measured by calculating posterior inclusion probabilities (PIPs). The PIPs quantify the relative importance of each exposure in the model, utilizing a scale from 0 to 1, where values nearer to 0 denote a lesser influence on the outcome. We used a threshold of 0.5 to determine the significance of the contribution. Moreover, both univariate and bivariate exposure-response functions were utilized to evaluate the individual effect and interaction of VOCs. In the BKMR model, we performed 10,000 iterations using the Markov Chain Monte Carlo method for all analyses.

Additionally, we analyzed the role of BMD in the relationship between VOC mixtures and growth indicators by conducting a mediation analysis. To effectively assess the growth and development of adolescents, we considered three representative measures of BMD (total BMD, subtotal BMD and lumbar spine BMD) to explore whether they mediated the association between VOC mixtures and growth indicators [55]. We initially employed three statistical models (GLM, WQS, and BKMR) to analyze the relationship between VOCs exposure and BMD. Subsequently, we used non-parametric bootstrapping ($n = 1000$) for the mediation analysis, assuming that the exposure-response relationship between VOCs (X) and growth indicators (Y) is mediated by BMD (M). This allowed us to explore the direct and indirect relationships as well as the degree of the mediating effect, and further evaluate the proportion of mediation (%) of BMD in the relationship between VOC mixtures and growth indicators.

We stratified the population by sex, race, PIR, smokers in the household, and physical activity to assess sensitivity and interactions in different subgroups. Besides, we conducted several sensitivity analyses to ensure the results of the relationship between VOCs and growth indicators. Firstly, we utilized principal component analysis (PCA) to evaluate the relationship between VOCs mixtures and growth indicators. By constructing an environmental risk score (ERS) using the first eight principal components and weighting them by variance contribution, which together explained over 80 % of the total variance [44]. Secondly, we used the baseline levels of urinary VOC metabolites as the exposure level without adjusting for urine creatinine, but included it as a covariate in fitting a GLM. Thirdly, we repeated the correlation analysis after excluding individuals with excessively dilute ($Cr < 30$ mg/dL) or concentrated ($Cr > 300$ mg/dL) urine samples. Fourthly, we included serum vitamin D levels as covariates in fitting the GLM. Finally, we employed the quantile g-computation (QGC) method to assess the relationship between VOCs mixtures and growth indicators, which combined the simplicity of weighted quantile regression with the flexibility of g-computation to estimate the overall exposure effect of increasing a quantile [56].

Finally, we conducted WQS regression to investigate the relationship

between VOCs mixtures and growth indicators in individuals with different serum vitamin D levels (vitamin D deficiency or vitamin D sufficiency). Weighted GLM was used to further explore the correlation between dietary vitamin D intake and BMD and growth indicators.

All statistical analyses were performed using R (version 4.3.1). Weighted GLM, LASSO regression, WQS regression, BKMR, PCA, QGC method, and mediation analyses were conducted using the R packages ‘survey’, ‘glmnet’, ‘gWQS’, ‘bkmr’, ‘FactoMineR’, ‘qgcomp’, and ‘mediation’, respectively. A two-sided P -value < 0.05 was considered statistically significant.

3. Results

3.1. Baseline demographic characteristics

The study included 746 participants with a median age of 15.0 years (quartile 1, quartile 3: 14.0, 18.0 years), and 50.5 % were male. Table 1 delineated the baseline characteristics of subjects. Most participants were Non-Hispanic White (26.0 %), engaged in moderate or vigorous physical activity (90.1 %), and had no family members who smoked (79.9 %). In addition, the average standing height and height-for-age of the entire adolescent sample were 164.8 ± 9.4 cm and 0.2 ± 1.1 , respectively. The median energy intake, weight, BMI, waist circumference, BMI-for-age, serum vitamin D and dietary vitamin D intake (quartile 1, quartile 3) were 1859.0 kcal (1398.8, 2533.5), 62.4 kg (53.0, 76.3), 22.6 kg/m² (19.8, 27.2), 78.2 cm (71.1, 89.0), 0.9 (0.0, 1.9), 4.0 nmol/L (1.3, 7.5) and 54.5 μ g (41.7, 65.7), respectively.

The distribution of VOC metabolites in urine was shown in Table S1. Among the 15 urinary VOC metabolites with a detection rate of > 80 %, the median concentration of DHBMA was the highest (345.00 ng/mL). And it was followed by 3HPMA (median concentration: 230.50 ng/mL). Fig. S4 showed the Pearson correlation coefficient between urinary VOC metabolite concentrations. In addition to 2MHA and ATCA, 3MHA+4MHA and ATCA, ATCA and CYMA, other VOC metabolites were remarkably correlated among themselves ($P < 0.05$).

Table 1
Baseline characteristics of US adolescents aged 12 to 19 years ($n = 746$).

	Mean (SD) or n (%) or Median (Q1, Q3)
Age (years)	15.0 (14.0, 18.0)
Sex	
Male	377 (50.5)
Female	369 (49.5)
Race	
Mexican American	167 (22.4)
Other Hispanic	70 (9.4)
Non-Hispanic White	194 (26.0)
Non-Hispanic Black	187 (25.0)
Other	128 (17.2)
Poverty income ratio	
< 1.0	226 (30.3)
1.0 –3.0	335 (44.9)
> 3.0	185 (24.8)
Physical activity	
Activity	672 (90.1)
Inactive	74 (9.9)
Household members that smoke	
Yes	150 (20.1)
No	596 (79.9)
Energy (kcal)	1859.0 (1398.8, 2533.5)
Standing height (cm)	164.8 (9.4)
Weight (kg)	62.4 (53.0, 76.3)
BMI (kg/m ²)	22.6 (19.8, 27.2)
Waist circumference (cm)	78.2 (71.1, 89.0)
Height-for-age	0.2 (1.1)
BMI-for-age	0.9 (0.0, 1.9)
Serum vitamin D (nmol/L)	4.0 (1.3, 7.5)
Dietary vitamin D intake (μ g)	54.5 (41.7, 65.7)

BMI, body mass index; SD, standard deviation.

3.2. Relationship of VOCs with growth indicators

The weighted GLM was used to analyze the relationship between 15 VOCs and growth indicators (standing height, weight, BMI, waist circumference, height-for-age and BMI-for-age). After adjusting for all confounding factors, we found that ATCA, DHBMA, 3HPMA, MA, MHBMA3 and PGA were negatively correlated with standing height. AAMA, ATCA, CYMA, DHBMA, 3HPMA, HPMMA, MA, MHBMA3 and PGA showed negatively association with weight. There was a significantly negatively connection between AAMA, DHBMA, HPMMA, MHBMA3, PGA and BMI. DHBMA, HPMMA, MHBMA3 and PGA were negatively correlated with waist circumference. AAMA, CYMA, DHBMA, HPMMA, MHBMA3 and PGA showed a significant negatively correlation with BMI-for-age (all P -value < 0.05). However, the GLM did not reveal a significant association between single urinary VOC and height-for-age (all P -value > 0.05) (Table 2 and Table S2).

The results of Pearson correlation showed a positive correlation between the concentrations of multiple VOCs in urine. To reduce the possible collinear variables in VOCs, we used an adaptive LASSO penalized regression model to screen VOCs that were stronger correlation with the growth indicators. Fig. 1 indicated the results of mean-squared error associated with the penalty parameter (λ). The optimal log-transformed λ values of standing height, weight, BMI, waist circumference, height-for-age, and BMI-for-age were -6.363 , -4.877 , -5.834 , -4.719 , -3.676 and -2.978 , respectively. For standing height, a total of 7 VOCs were selected, such as 3MHA+ 4MHA, AMCC, ATCA, BMA, CEMA, CYMA and PGA. For weight, 3MHA+ 4MHA, ATCA, CEMA, CYMA, DHBMA, HPMMA, MHBMA3 and PAG were included in the analysis. For BMI, the results included the remaining 12 VOCs in addition to 2MHA, AAMA, and 2HPMA. For waist circumference, a total of 3MHA+ 4MHA, ATCA, HPMMA and PAG were screened. For height-for-age, the analysis included 3MHA+ 4MHA, CEMA, CYMA, DHBMA, HPMMA, and MA. For BMI-for-age, the results included 5 VOCs such as

3MHA+ 4MHA, BMA, DHBMA, HPMMA and PGA (Fig. S5).

To assess the mixed effects of VOCs on growth indicators, we performed WQS regression analysis on the variables screened by LASSO regression. After adjusting for potential confounders, we found that WQS indices were significantly negatively associated with standing height (β : -0.006 , 95 % CI: -0.008 , -0.004), weight (β : -0.023 , 95 % CI: -0.037 , -0.010), BMI (β : -0.016 , 95 % CI: -0.027 , -0.004), waist circumference (β : -0.010 , 95 % CI: -0.018 , -0.002), and BMI-for-age (β : -0.084 , 95 % CI: -0.147 , -0.021) (Table S3). As shown in Fig. 2, for weight, BMI, and BMI-for-age, DHBMA was the highest WQS weight among various VOCs. However, for standing height and waist circumference, the highest contributions were made by ATCA (0.465) and PGA (0.386), respectively.

After the suitable variables were selected by LASSO regression, the effects of VOCs mixture on growth indicators were investigated by BKMR model. As shown in Fig. 3, when fixed at the median concentration with VOCs, VOCs mixture concentration was negatively correlated with standing height, weight, BMI, waist circumference, and BMI-for-age. Table S4 summarizes the PIP results of each VOCs in the BKMR model. HPMMA contributed the most to the overall association of VOCs mixtures with weight (0.858), BMI (0.872), and BMI-for-age (0.994). However, for the overall association between VOCs mixtures and standing height and waist circumference, none of the VOCs had PIP greater than 0.5. To determine the effect of a single VOC on growth indicators, we fixed the concentrations of other VOCs at median levels and observed the univariate exposure-response relationship function of VOCs. As shown in Fig. S6, HPMMA had negative nonlinear effects on weight, BMI and BMI-for-age. However, DHBMA had a positive linear effect on height-for-age. In addition, Fig. S7 showed the relationship between individual VOCs and growth indicators when remaining the other VOCs at the 25th, 50th, and 75th percentiles, respectively. We observed that HPMMA was negatively correlated with waist circumference and BMI-for-age, and ATCA was negatively associated with

Table 2

Associations between single urinary VOC metabolites and growth indicators in US adolescents ^a.

VOCs	Standing height β (95 % CI)	Weight β (95 % CI)	BMI β (95 % CI)	Waist circumference β (95 % CI)	Height-for-age β (95 % CI)	BMI-for-age β (95 % CI)
2MHA	$-0.001(-0.006, 0.004)$	$-0.021(-0.046, 0.004)$	$-0.018(-0.042, 0.005)$	$-0.012(-0.027, 0.003)$	$-0.024(-0.120, 0.072)$	$-0.110(-0.258, 0.037)$
3MHA+ 4MHA	$-1.496e-04(-0.005, 0.005)$	$-0.014(-0.042, 0.015)$	$-0.013(-0.040, 0.014)$	$-0.007(-0.025, 0.012)$	$-0.008(-0.104, 0.088)$	$-0.079(-0.249, 0.091)$
AAMA	$-0.003(-0.009, 0.003)$	$-0.048(-0.088, -0.009)$	$-0.042(-0.076, -0.007)$	$-0.024(-0.050, 0.002)$	$-0.028(-0.178, 0.122)$	$-0.273(-0.489, -0.056)$
AMCC	$-0.005(-0.013, 0.002)$	$-0.015(-0.048, 0.018)$	$-0.004(-0.032, 0.024)$	$0.004(-0.020, 0.028)$	$-0.041(-0.215, 0.134)$	$-0.026(-0.199, 0.147)$
ATCA	$-0.010(-0.015, -0.006)$	$-0.034(-0.063, -0.004)$	$-0.013(-0.042, 0.016)$	$-0.009(-0.027, 0.010)$	$-0.079(-0.179, 0.021)$	$-0.040(-0.216, 0.135)$
BMA	$-0.006(-0.012, 1.063e-04)$	$-0.006(-0.040, 0.028)$	$0.006(-0.024, 0.036)$	$0.001(-0.022, 0.025)$	$-0.089(-0.221, 0.043)$	$0.039(-0.141, 0.219)$
CEMA	$-0.006(-0.013, 0.001)$	$-0.031(-0.067, 0.005)$	$-0.019(-0.049, 0.010)$	$-0.007(-0.031, 0.016)$	$-0.124(-0.294, 0.047)$	$-0.134(-0.319, 0.051)$
CYMA	$-0.002(-0.004, 0.001)$	$-0.018(-0.034, -0.001)$	$-0.014(-0.029, 0.001)$	$-0.008(-0.020, 0.003)$	$-0.035(-0.109, 0.039)$	$-0.092(-0.183, -0.001)$
DHBMA	$-0.013(-0.025, -0.001)$	$-0.108(-0.169, -0.047)$	$-0.083(-0.135, -0.031)$	$-0.043(-0.082, -0.003)$	$-0.065(-0.324, 0.194)$	$-0.497(-0.829, -0.164)$
2HPMA	$-0.003(-0.007, 0.001)$	$-0.025(-0.051, 0.001)$	$-0.019(-0.043, 0.005)$	$-0.015(-0.038, 0.007)$	$-0.021(-0.130, 0.089)$	$-0.104(-0.248, 0.040)$
3HPMA	$-0.008(-0.014, -0.001)$	$-0.044(-0.078, -0.010)$	$-0.029(-0.059, 0.002)$	$-0.017(-0.039, 0.006)$	$-0.153(-0.306, 1.021e-04)$	$-0.179(-0.366, 0.008)$
HPMMA	$-0.006(-0.015, 0.003)$	$-0.074(-0.104, -0.043)$	$-0.062(-0.088, -0.036)$	$-0.035(-0.055, -0.016)$	$-0.099(-0.307, 0.108)$	$-0.370(-0.540, -0.201)$
MA	$-0.012(-0.021, -0.003)$	$-0.051(-0.096, -0.006)$	$-0.027(-0.062, 0.007)$	$-0.021(-0.046, 0.005)$	$-0.166(-0.360, 0.029)$	$-0.168(-0.377, 0.040)$
MHBMA3	$-0.006(-0.012, -0.001)$	$-0.048(-0.079, -0.018)$	$-0.036(-0.061, -0.011)$	$-0.021(-0.041, -0.002)$	$-0.107(-0.235, 0.020)$	$-0.221(-0.375, -0.067)$
PGA	$-0.010(-0.020, -0.001)$	$-0.062(-0.105, -0.020)$	$-0.042(-0.077, -0.006)$	$-0.031(-0.059, -0.002)$	$-0.093(-0.304, 0.118)$	$-0.218(-0.433, -0.003)$

^a Models were adjusted for age, sex, race, poverty income ratio, smoker/s in the household, physical activity and energy. BMI, body mass index; CI, confidence interval; VOCs, volatile organic compounds.

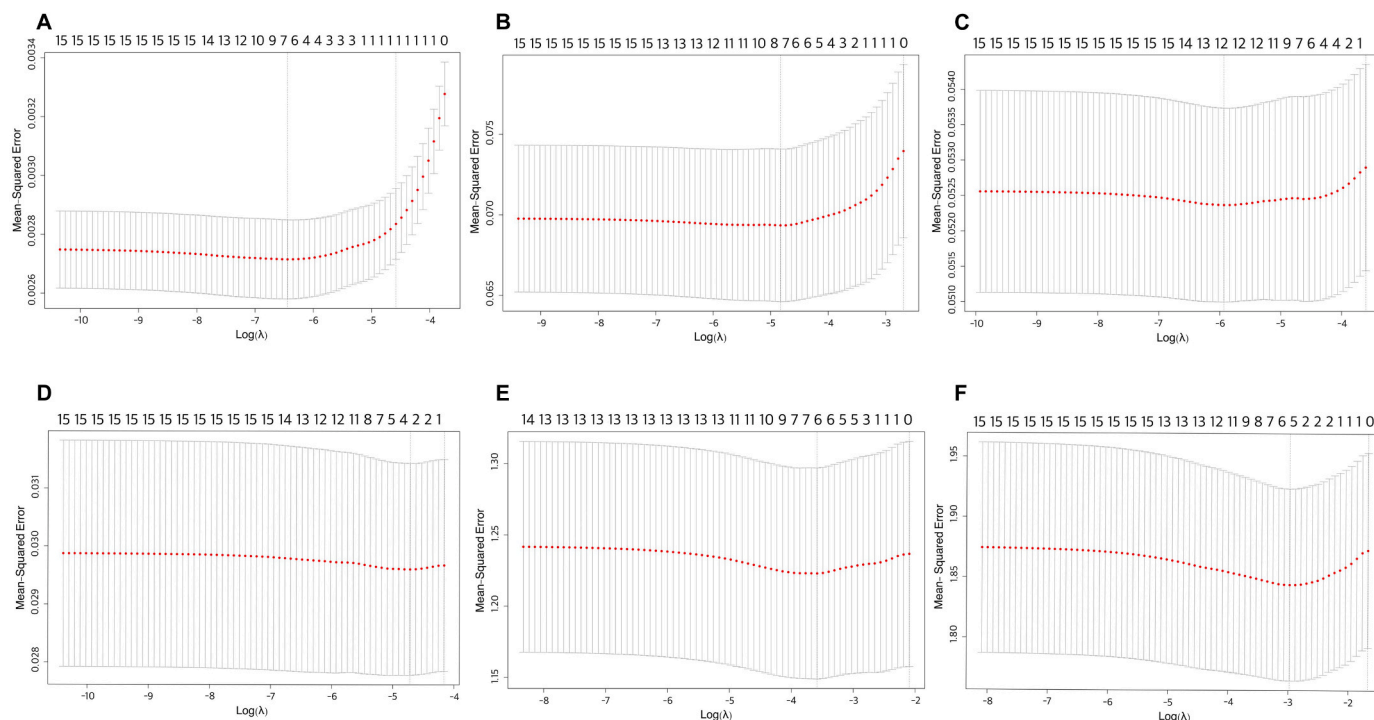


Fig. 1. 10-fold cross-validation curves between log-transformed LASSO penalty parameter (λ) and standing height (A), weight (B), BMI (C), waist circumference (D), Height-for-age (E), BMI-for-age (F) in US adolescents. The independent variables are ln-transformed 15 VOCs. The left black dotted line is optimal value of λ at which the minimum mean square error is calculated, while the right black dotted line is the value of λ in the simplest model obtained at one standard error of the minimum mean square error. BMI, body mass index; LASSO, least absolute shrinkage and selection operator; VOCs, volatile organic compounds.

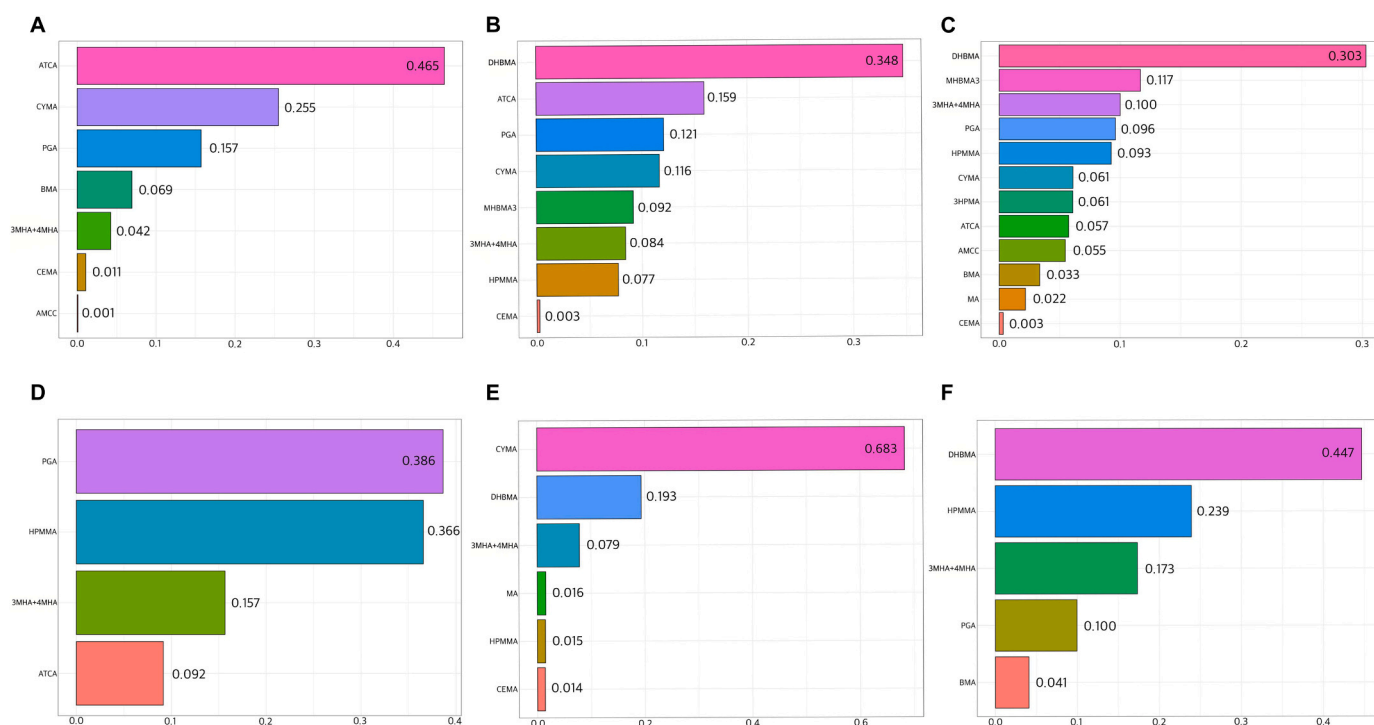


Fig. 2. The WQS model weights of screened urinary VOCs on standing height (A), weight (B), BMI (C), waist circumference (D), Height-for-age (E), and BMI-for-age (F) in negative direction. Models were adjusted for age, sex, race, poverty income ratio, smoker/s in the household, physical activity, and energy. BMI, body mass index; VOCs, volatile organic compounds; WQS, weighted quantile sum.

standing height and weight. Finally, we explored the bivariate exposure-response function of a single VOC for the second VOCs at the 25th, 50th, and 75th percentiles by fixing the other VOC concentrations at the

median level. The results showed that there were significant interaction effects between BMA and 3MHA+ 4MHA, DHBA and 3MHA+ 4MHA, and HPMMA and 3MHA+ 4MHA for BMI-for-age (Fig. S8).

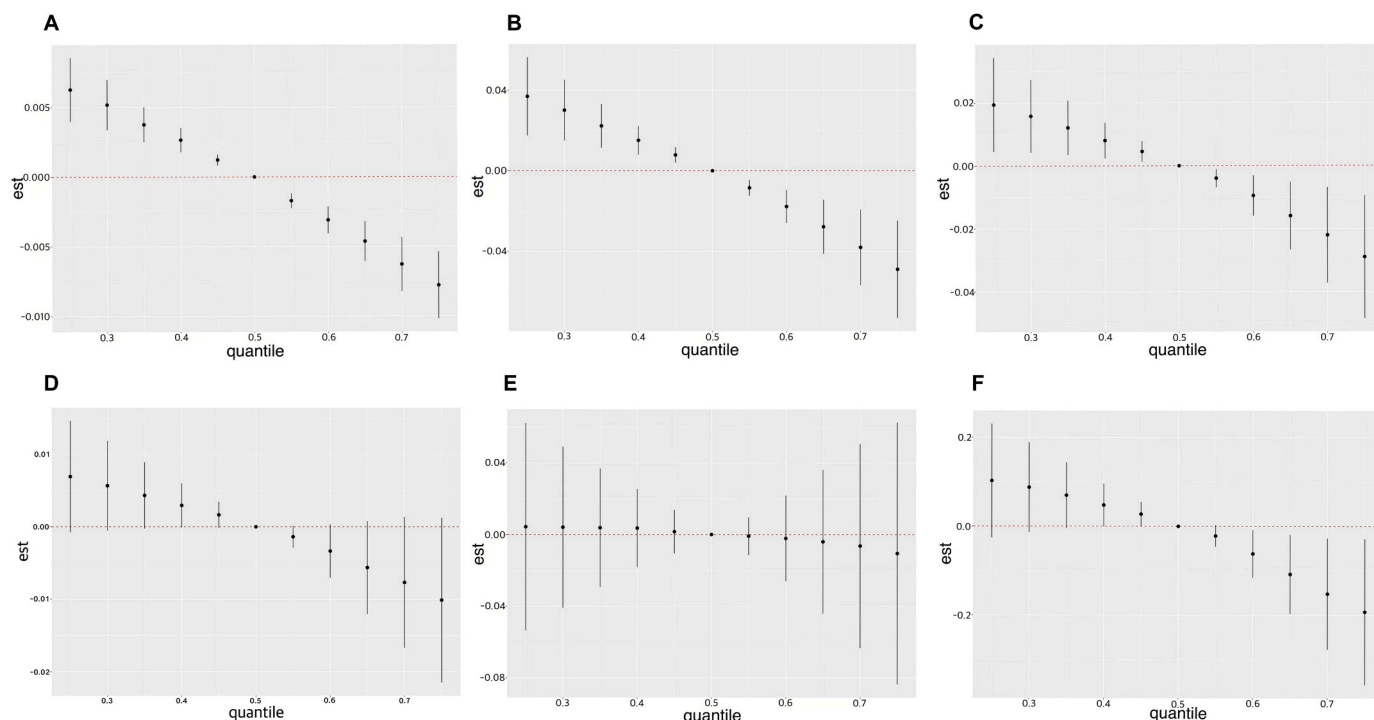


Fig. 3. The relationship between urinary VOCs mixture and standing height (A), weight (B), BMI (C), waist circumference (D), Height-for-age (E), and BMI-for-age (F) in BKMR model. Models were adjusted for age, sex, race, poverty income ratio, smoker/s in the household, physical activity, and energy. BMI, body mass index; BKMR, Bayesian kernel machine regression; VOCs, volatile organic compounds.

3.3. Associations between VOCs and BMD

The relationship between 15 VOCs and BMD was shown in Table S5. After adjusting for all potential confounders, AAMA, ATCA, CEMA, DHBMA, 3HPMA, HPMMA, MA, MHBMA3 and PGA were negatively correlated with the three types of BMD.

Fig. S9 indicated the results of 15 VOCs associated with BMD in LASSO regression model. The optimal log-transformed λ values of total BMD, subtotal BMD, and lumbar spine BMD were -2.866 , -2.872 and -2.790 , respectively. For total BMD, 14 VOCs excluding 3HPMA were included in the analysis. For subtotal BMD, there were 13 VOCs including 2MHA, 3MHA+ 4MHA, AAMA, AMCC, ATCA, BMA, CYMA, DHBMA, 2HPMA, HPMMA, MA, MHBMA3 and PGA. For lumbar spine BMD, 13 VOCs except 3HPMA and HPMMA were screened.

The relationship between mixed VOCs and BMD was analyzed by constructing WQS regression model and BKMR model based on the key variables selected by LASSO. As shown in Table S6, WQS index was negatively correlated with BMD. ATCA was the highest WQS weigh among various VOCs for total BMD (0.339), subtotal BMD (0.369), and lumbar spine BMD (0.393). The BKMR model further demonstrated the negative correlation between mixed VOCs and BMD (Fig. S10). And Table S7 summarized the PIP results of each VOCs in the BKMR model. ATCA contributed the most to the overall association of VOCs mixtures with BMD.

3.4. Associations between BMD and growth indicators

After adjusting for all confounding factors, total BMD was found to be positively associated with standing height (β : 0.169, 95 % CI: 0.134, 0.205), weight (β : 1.207, 95 % CI: 1.027, 1.387), BMI (β : 0.868, 95 % CI: 0.681, 1.055), waist circumference (β : 0.530, 95 % CI: 0.403, 0.658), height-for-age (β : 2.534, 95 % CI: 1.732, 3.337) and BMI-for-age (β : 5.241, 95 % CI: 4.006, 6.477). Similarly, this positive correlation also existed between subtotal BMD, lumbar spine BMD, and growth indicators (all P -value < 0.05) (Table S8).

3.5. BMD mediated the relationship between VOCs and growth indicators

We further evaluated whether BMD mediated the negative relationship between VOCs mixtures and growth indicators. And we found that the mediated proportion (%) of total BMD in the relationship between VOCs mixture and standing height, weight, BMI, waist circumference, and BMI-for-age were 23.7 %, 31.3 %, 34.1 %, 23.4 %, and 31.2 %, respectively. The mediated proportion (%) of the subtotal BMD in the relationship between VOCs mixture and standing height, weight, BMI, waist circumference and BMI-for-age were 53.4 %, 48.9 %, 46.5 %, 38.8 %, and 39.0 %, respectively. As for standing height, weight, BMI, waist circumference and BMI-for-age, the mediated proportion of lumbar spine BMD was 18.1 %, 11.9 %, 10.1 %, 4.3 %, and 11.3 % for VOCs mixture, respectively. However, BMD had no significant mediating effect on the association between VOCs and height-for-age (Figs. 4–6 and Table S9).

3.6. Subgroup analysis

As presented in Supplementary Table S10–S14, we stratified the analysis by sex, race, PIR, smokers in the household, and physical activity. Results showed that race, PIR, smokers in the household, and physical activity did not change the association between VOCs and growth indicators. Interestingly, sex significantly changed the association between VOCs and growth indicators (P for interaction < 0.05). VOCs were more negatively correlated with standing height, weight, BMI, height-for-age, and BMI-for-age in male (Table S10–S14).

3.7. Sensitivity analysis

To assess the robustness of the association between VOCs mixtures and adolescent growth, we first performed PCA analyses for 15 VOCs, which were generally consistent with LASSO regression results. There was a significant negative correlation between PCA based ERS and growth indicators (Table S15). Secondly, the concentration of VOC

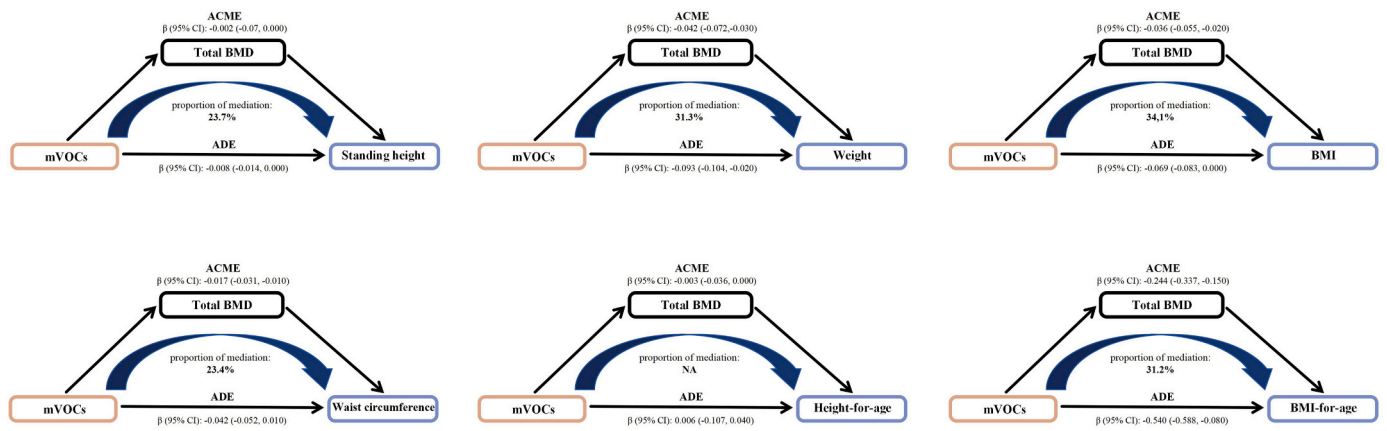


Fig. 4. Mediation effects of total BMD in the associations of urinary VOCs with growth indicators. BMD, bone mineral density; BMI, body mass index; mVOCs, metabolites of volatile organic compounds.

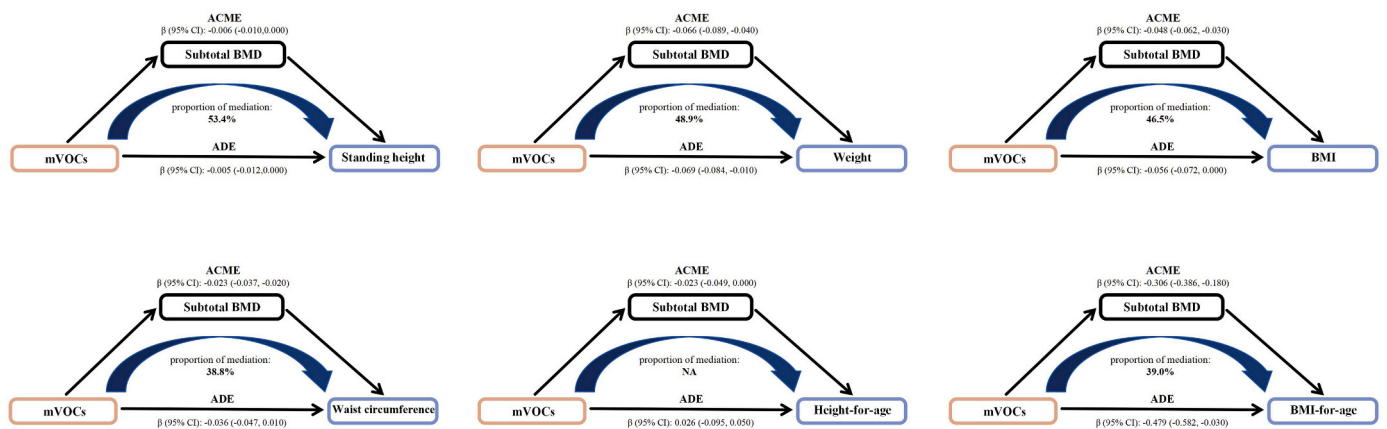


Fig. 5. Mediation effects of subtotal BMD in the associations of urinary VOCs with growth indicators. BMD, bone mineral density; BMI, body mass index; mVOCs, metabolites of volatile organic compounds.

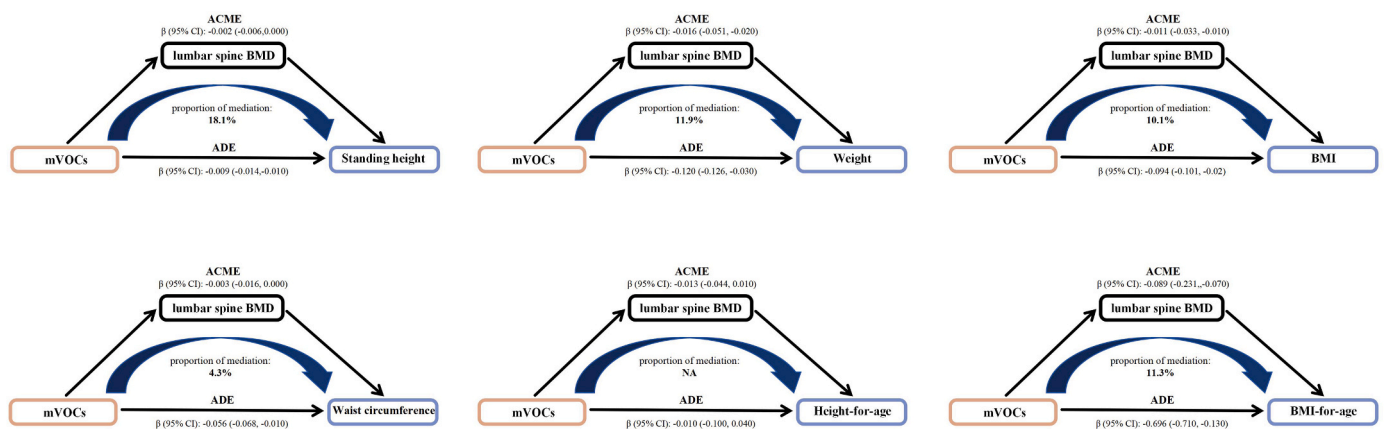


Fig. 6. Mediation effects of lumbar spine BMD in the associations of urinary VOCs with growth indicators. BMD, bone mineral density; BMI, body mass index; mVOCs, metabolites of volatile organic compounds.

metabolites in urine was used as exposures without adjusting for urinary creatinine, but as a covariate for fitting GLM, and its association with growth indicators was similar to the previous analysis (Table S16). In addition, after excluding samples from individuals whose urine was too dilute ($Cr < 30$ mg/dL) or too dense ($Cr > 300$ mg/dL), the association was consistent with the previous results (Table S17). Besides, when serum vitamin D was considered as a covariable, the association

between VOCs and growth indicators remained robust (Table S18). Finally, QGC analysis also showed a significant negative correlation between VOCs mixture and growth indicators (Table S19).

3.8. Information on mechanisms and targets

To further investigate the mechanism of the effect of mixed exposure

to VOCs on growth indicators, we conducted functional enrichment analysis and PPI network analysis on the intersection of VOCs targets and adolescent growth and development related proteins. From SEA and PharmMapper databases, 544 target proteins of VOCs were obtained through the intersection of the results from the two databases (Fig. 7A). The database platforms of Genecards were used to search for the approved targets that affected adolescent growth and development. 1484 proteins related to adolescent growth and development were acquired from GeneCards database. We screened 132 overlapping targets as prospective targets linked to growth retardation caused by VOCs by intersecting the targets of VOCs with proteins related to adolescent growth and development (Fig. 7B). GO analysis was performed to identify prospective targets participating in biological processes (BPs), cellular components (CCs), and molecular functions (MFs). The GO analysis results showed that GO terms, such as response to xenobiotic stimulus (GO:0009410), negative regulation of apoptotic process (GO:0043066), response to hypoxia (GO:0001666), cytoplasm (GO:0005737), cell surface (GO:0009986), macromolecular complex (GO:0032991), enzyme binding (GO:0032991), identical protein binding (GO:0042802), and endopeptidase activity (GO:0004175) were enriched (Fig. 7C; Table S20). In the KEGG enrichment analysis, the top five were Pathways in cancer, Lipid and atherosclerosis, Proteoglycans in cancer, IL-17 signaling pathway, Fluid shear stress and atherosclerosis (Fig. 7D; Table S21). We also established a PPI network with the overlapping targets through the STRING database. As shown in Fig. 7E, the PPI network had 132 nodes and 1442 edges. Each protein was represented by a node, while each edge presented one interaction. The essentiality of overlapping targets to the interacting network was further evaluated through MCC, MNC, degree, EPC, closeness, radiality, betweenness, stress, and BottleNeck algorithms. Based on the above algorithms, the overlapping targets were scored and sorted, and the top 10 scoring proteins of each algorithm was considered as candidate targets of VOCs. We sought to further predict the potential targets of VOCs by taking the intersection of candidate targets from nine algorithms. The identified hub proteins include IL-1 β , CASP3, HSP90AA1, SRC, and FOS (Fig. 7F; Table S22). Therefore, it can be inferred that IL-17 signaling pathway might be a potential mechanism for the association of VOCs and adolescent growth and development, and IL-1 β , CASP3, HSP90AA1, SRC, and FOS proteins were possible target of VOCs.

3.9. The effect of vitamin D on the relationships of VOCs with BMD and growth indicators

To identify potential intervention strategies for the adverse effects of VOCs exposure on BMD and adolescent growth, we first analyzed the effects of serum vitamin D levels on the relationships of VOCs with BMD and growth indicators. Table 3 indicated that the WQS index of VOCs was negatively associated with total BMD (β : -0.017, 95 % CI: -0.026, -0.007), subtotal BMD (β : -0.016, 95 % CI: -0.025, -0.007), lumbar spine BMD (β : -0.021, 95 % CI: -0.032, -0.010), standing height (β : -0.005, 95 % CI: -0.009, -0.002), weight (β : -0.045, 95 % CI: -0.069, -0.020), BMI (β : -0.034, 95 % CI: -0.056, -0.012), waist circumference (β : -0.022, 95 % CI: -0.039, -0.005), and BMI-for-age (β : -0.214, 95 % CI: -0.339, -0.090) in vitamin D deficiency adolescents. However, for individuals with vitamin D sufficiency, the WQS index of VOCs was only negatively correlation with BMD, standing height, and weight. Second, we also investigated the relationship between dietary vitamin D and BMD and growth indicators, and the results showed that dietary vitamin D intake was significantly positively correlated with total BMD, subtotal BMD, standing height, weight, BMI, waist circumference, and BMI-for-age (all P -value < 0.05) (Fig. S11).

4. Discussion

In this cross-sectional study of US adolescents, we found a significant negative association between VOCs exposure and growth indicators

(standing height, weight, BMI, waist circumference, and BMI-for-age), and the BMD levels mediated the association between VOCs mixture and growth indicators. The results of functional enrichment analysis suggested that VOCs may be related to adolescent growth and development through underlying biological mechanisms, and the most important module of this result was significantly related to the IL-17 signaling pathway. In addition, VOCs may also act on the growth and development of adolescents by targeting proteins such as IL-1 β , CASP 3, HSP90AA1, SRC and FOS. The association between dietary vitamin D and BMD and growth indicators indicated that dietary vitamin D intake may be a feasible strategy to prevent the adverse effects of VOCs exposure on adolescent growth.

In recent years, with the rapid development of industrialization, environmental pollution has become one of the main causes of human diseases [57]. Existing findings suggested that exposure to air pollutants may delay growth in children and adolescents. A Polish study showed that the average stature of children and adolescents decreased as the level of air pollution in the place where they live increased [58]. The growth rate of children height living in areas with high pollution was reduced by an average of 1.5 cm compared with children living in areas with low pollution [59]. As widely exposed air pollutants in daily life, VOCs come from a variety of sources (such as tobacco smoke, building materials, furniture, and cosmetics) [60]. With the continuous improvement of outdoor air quality, the adverse effects of indoor air pollution on health have attracted wide attention. Studies showed that people in developed countries spent up to 90 % of their time indoors [61]. As the most common indoor air pollutant, the health risks of VOCs cannot be ignored. Thus, a comprehensive assessment of the harm of VOCs exposure to adolescent growth was of great public health significance. To our knowledge, this is the first study to evaluate both single and overall associations between VOCs exposure and adolescent growth, while providing new insights into the mechanisms by which VOCs exposure affects adolescent growth. In addition, our findings may fill gaps in the understanding of the hazards of exposure to VOCs to human health.

At present, there are few studies on VOCs exposure and growth. An epidemiological study conducted in the US showed that exposed to trichloroethylene was associated with low birth weight and fetal growth restriction [62]. The results of another study showed that the exposure of prenatal VOCs was associated with a reduction in neonatal weight [63]. An experimental animal study also demonstrated that prenatal exposure to styrene had effect on the growth and development of their offspring [64]. Current research mostly focuses on single VOC components and the adverse effects of VOCs on newborns, and there is currently a lack of research evidence on the impact of VOCs exposure with adolescent growth and development. In our study, for a single VOC, we found that 1, 3-butadiene metabolites (DHBMA and MHBMA3), ethylbenzene and styrene metabolite (PGA) were negatively associated with growth indicators (standing height, weight, BMI, waist circumference, BMI-for-age). In addition, we also found that DHBMA caused the highest degree of adverse effects on growth indicators, which may be related to the highest concentration it exhibited in all participants. In stratified analysis, sex markedly modified the association between VOCs and growth indicators (P for interaction < 0.05). The results showed that male growth and development during adolescence were more prone to be affected by VOCs exposure than female. Although the exact mechanism is still unclear, we can consider several possible hypotheses. Firstly, the female immune system is evolutionarily capable of responding more forcefully to the threat of infection [65]. Casimir et al. [66] found that males produced more inflammatory factors after being stimulated by mitogen monocytes. Secondly, males undergo later pubertal development than females [67]. Therefore, exposure to VOCs in male adolescents deserves special attention.

Considering that humans are usually exposed to mixtures of VOCs in their daily lives, research that only focuses on a single VOC component may cause some degree of bias [68]. Therefore, we conducted a

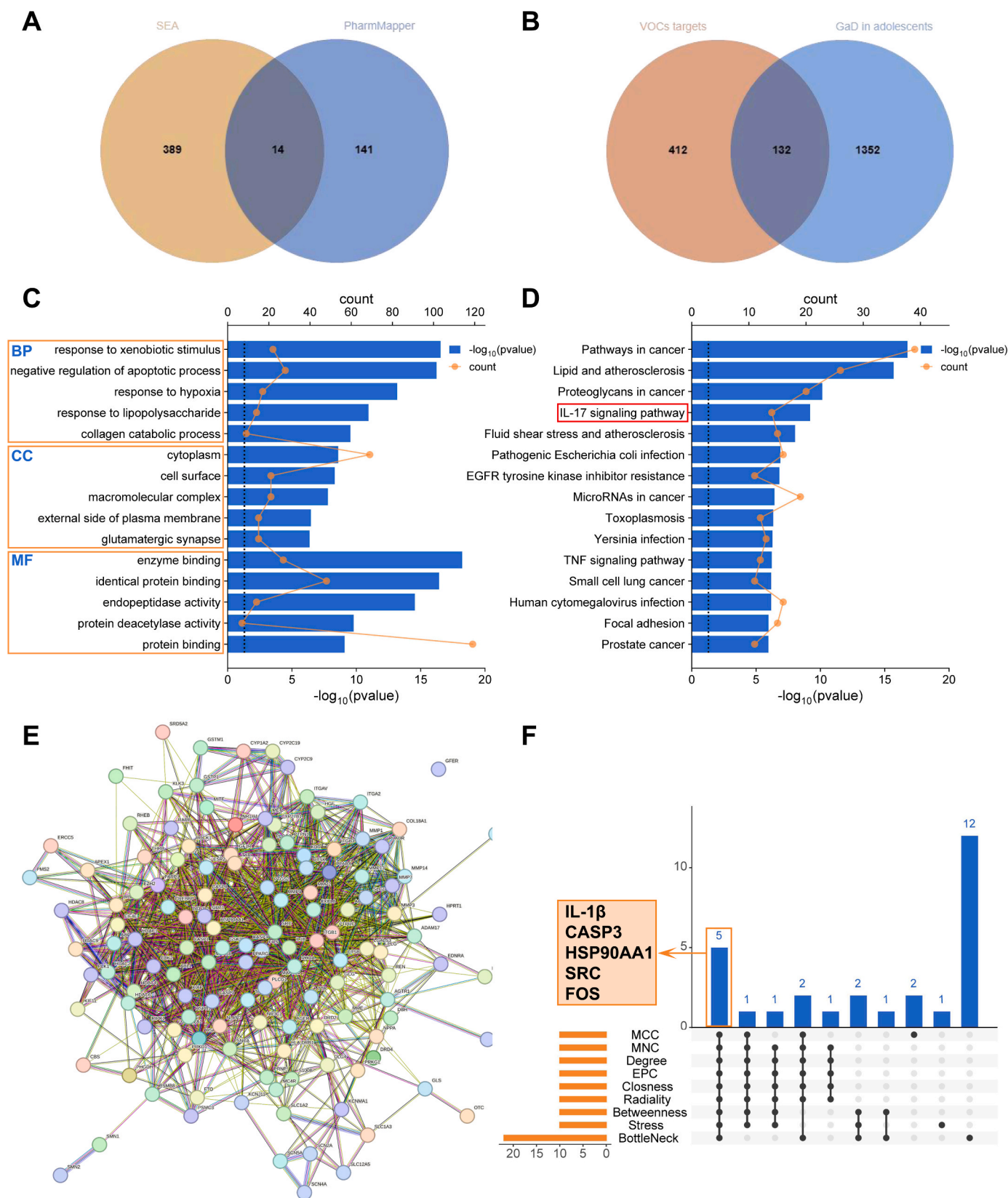


Fig. 7. The potential targets and mechanisms of VOCs were predicted through network pharmacological method. (A) Venn graph showing the numbers of predicted VOCs targets. (B) The overlapping targets of VOCs targets and genes related to adolescent growth and development. (C) Bar plot of GO enrichment analysis. (D) Bar plot of KEGG enrichment analysis. (E) The PPI network of overlapping targets. (F) UpSet plot of 9 topological algorithms. BP, biological processes; CC, cellular component; GaD, growth and development; GO, Gene Ontology; KEGG, Kyoto Encyclopedia of Genes and Genomes; MF, molecular function; PPI, protein-protein interaction; SEA, Similarity Ensemble Approach; VOCs, volatile organic compounds.

Table 3

Associations of WQS regression index with BMD and growth indicators stratified by vitamin D levels in US adolescents ^a.

	Vitamin D deficiency		Vitamin D sufficiency	
	β (95 % CI)	P-value	β (95 % CI)	P-value
BMD				
Total BMD	-0.017 (-0.026, -0.007)	< 0.001	-0.016 (-0.024, -0.009)	< 0.001
Subtotal BMD	-0.016 (-0.025, -0.007)	< 0.001	-0.018 (-0.026, -0.010)	< 0.001
Lumbar spine BMD	-0.021 (-0.032, -0.010)	< 0.001	-0.019 (-0.028, -0.009)	< 0.001
Growth indicators				
Standing height	-0.005 (-0.009, -0.002)	0.005	-0.009 (-0.012, -0.006)	< 0.001
Weight	-0.045 (-0.069, -0.020)	< 0.001	-0.024 (-0.040, -0.005)	0.008
BMI	-0.034 (-0.056, -0.012)	0.003	-0.010 (-0.022, 0.007)	0.162
Waist circumference	-0.022 (-0.039, -0.005)	0.012	-0.006 (-0.014, 0.004)	0.219
BMI-for-age	-0.214 (-0.339, -0.090)	< 0.001	-0.048 (-0.133, 0.031)	0.285

^a Models were adjusted for age, sex, race, poverty income ratio, smoker/s in the household, physical activity and energy. BMD, bone mineral density; BMI, body mass index; CI, confidence interval; WQS, weighted quantile sum.

comprehensive analysis to further explore the combined impact of VOCs exposure on adolescent growth by using WQS and BKMR models, so as to prove the potential adverse effects of VOCs on adolescent growth. Both models consistently indicated combined VOCs exposure was inversely associated with standing height, weight, BMI, waist circumference, and BMI-for-age. In addition, identifying key toxic VOCs compounds has important implications for promoting health. Interestingly, in the WQS model, the 1, 3-butadiene metabolite DHBMA was the main contributor to the adverse effects of VOCs on growth indicators (weight, BMI, and BMI-for-age). 1, 3-butadiene is ubiquitous in the environment. Due to the significant potential threat to human health, the Occupational Safety and Health Administration has recommended that occupational exposure standards for this chemical be lowered [69]. In the BKMR model, the DHBMA remained the main contributor to the adverse effects of VOCs on BMI-for-age. However, crotonaldehyde metabolite HPMMA was a major contributor to more affected growth indicators, such as weight, BMI, and BMI-for-age. The reason for this difference is mainly attributed to the different estimation methods of variable importance. In addition, in the WQS model, HPMMA also ranked prominently in the ranking of weighted contribution. At the same time, our study found that both DHBMA and HPMMA were at relatively high levels in the urine of US adolescents. Therefore, those adverse effects on the growth of adolescents deserve attention. Our research findings emphasized the necessity of implementing interventions to reduce VOCs exposure during adolescence, such as increasing indoor ventilation, the selection of low-emitting products and installing an air purifier.

In this study, we found that VOCs were negatively correlated with BMD, and BMD was positively correlated with growth indicators. Similar to our results, a cross-sectional study of NHANES suggested that long-term exposure to chloroform may lead to a reduction in BMD [28]. Zanlorenzi et al.'s study showed that BMD was directly related to growth indicators [70]. Based on the above research results, we surmise that BMD might play a significant mediation effect in the associations of exposure to VOCs with growth indicators. The results showed that BMD had a significant mediating role in the association between VOCs exposure and adolescent growth, and the mediated proportion ranging from 4.3 % to 53.4 %. However, the mechanism by which BMD mediates the adverse effects of VOCs exposure on growth remains unclear. Therefore, we conducted GO and KEGG enrichment analyses for the overlapping targets between VOCs targets and adolescent growth and

development proteins. The results showed overlapping targets were significantly enriched in IL-17 signaling pathways. Previous studies have shown that the IL-17 signaling pathway is closely related to bone health and growth and development in children and adolescents [71–73]. IL-17, as a pleiotropic factor produced by Th17 T cells, can induce excess inflammation through cytokine cascade [74]. It has been reported that treatment with anti-IL-17 antibodies or silencing the IL-17 signaling pathway can reduce joint inflammation and articular bone erosion [75, 76]. This result may be achieved by reducing bone loss by inhibiting RANKL expression on osteoblasts or stimulating the up-regulation of regulatory cytokines such as IL-4, IL-12 and gamma-interferon. Also, the increase of inflammatory factors also has adverse effects on growth and development [77]. One study in newborns showed that IL-17 may affect pre-eclampsia related fetal growth restriction [73]. Another study also showed that air pollutants may lead to osteoclast production by inducing inflammatory factors [78]. Therefore, VOCs exposure may affect BMD by activating the IL-17 signaling pathway, thereby having adverse effects on the growth and development of adolescents.

Our findings also revealed the role of dietary vitamin D on BMD and growth indicators. VOCs exposure has been reported to be associated with a decrease in BMD [30]. At the same time, multiple studies have shown that exposure to air pollutants during pregnancy is detrimental to fetal growth [79,80]. For children and adolescents, several epidemiological studies have shown that vitamin D intake is beneficial for their growth and bone development [36,81]. Interestingly, there were also several studies demonstrating that vitamin D supplementation had no effect on the bone health and growth and development of children and adolescents [82–84]. Our results showed that the adverse effects of VOCs on adolescents with low serum vitamin D levels were significantly greater than those with high serum vitamin D levels. This result suggested that dietary intake of vitamin D may alleviate the impact of VOCs on adolescent growth. Therefore, we further investigated the effect of dietary vitamin D intake on growth indicators in adolescents, and found that there was a positive correlation between vitamin D intake and BMD and growth indicators in adolescents. The above results further emphasized the beneficial effects of dietary supplementation of vitamin D on the growth and development of adolescents, especially those with high levels of VOCs in urine. In addition to obtaining vitamin D through dietary sources, sun exposure has long been considered essential for maintaining a sufficient vitamin D status [85]. Therefore, supplementing with vitamin D and increasing their basking activity may be two effective ways to prevent the adverse effects of VOCs exposure on adolescent growth and development.

Our study has several advantages. Firstly, this is the first study to comprehensively explore the effects of single and mixed VOCs exposure on adolescent growth. At the same time, we found the mediating role of BMD in VOCs exposure and adolescent growth. This provided more comprehensive evidence of the adverse effects of VOCs on adolescent growth. Secondly, we further explored the potential mechanism of the adverse effects of VOCs on adolescent growth by using the method of network pharmacology analysis, and found that VOCs may lead to the activation of IL-17 signaling pathway. In addition, we also found that dietary supplementation of vitamin D may be a feasible way to prevent the adverse effects of VOCs exposure on growth. Finally, we performed various stratification analyses and sensitivity analyses to demonstrate the robustness of our findings. Nonetheless, there are some limitations to this study. First, this is a cross-sectional study, and we can't make causal inferences about the relationship between VOCs and adolescent growth. Second, despite adjusting for some covariates, we still cannot exclude the possibility of other potentially unmeasured confounders having an impact on this study, such as the participants' occupational exposure, disease states and medication use. Thirdly, single-point urinary samples from NHANES are inherently difficult to assess long-term VOCs exposure levels, and results from 24-hour urine samples may be more reliable. Fourthly, the accuracy of VOC exposure assessment may be influenced by individual differences, the time of sample collection, storage time,

and storage temperature. Finally, assessing the effectiveness of vitamin D in preventing the adverse effects of VOCs on adolescent growth still needs to be further investigated in cells, animals, and populations.

5. Conclusion

In summary, our results suggested that exposure to single VOC and mixed VOCs may affect adolescent growth through three statistical models. 1, 3-butadiene and crotonaldehyde were identified as the main contributors due to their higher weights and the PIPs. The BMD played an important mediating role in the adverse effects of VOCs on adolescent growth. According to the results of PPI and functional enrichment analysis, we believed that the IL-17 signaling pathway may be the potential mechanism of VOCs exposure affecting adolescent growth. Additionally, the attenuation of this association by elevated serum vitamin D and the positive correlation between vitamin D and growth indicators indicated that dietary supplementation of vitamin D may be a viable strategy to prevent the adverse effects of VOCs exposure on adolescent growth.

Environmental implication

Volatile organic compounds (VOCs), a group of pollutants pervasive in daily life, are absorbed by the human body in various ways, resulting in the occurrence of various diseases. The results of this study suggested that VOCs exposure may affect adolescent growth, with bone mineral density playing an important mediating role, and IL-17 signaling pathway was the potential mechanism. Vitamin D may be a potential intervention strategy to prevent the adverse effects of VOCs on adolescent growth. Our results not only revealed the potential relationship and mechanism between VOCs exposure and adolescent growth, but also provided a potential intervention strategy.

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CRediT authorship contribution statement

Jingyi Ren: Conceptualization, Methodology, Validation, Writing-original draft. **Xiaoya Sun:** Data curation, Formal analysis, Validation, Writing-original draft. **Zhenao Zhang:** Data curation, Formal analysis, Writing-original draft. **Huanting Pei:** Formal analysis. **Yadong Zhang:** Formal analysis. **Rui Wen:** Data curation. **Simeng Qiao:** Data curation. **Zidan Wang:** Investigation. **Weican Zhang:** Investigation. **Jinshi Zuo:** Investigation. **Yuxia Ma:** Funding acquisition, Writing-review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jhazmat.2024.135422](https://doi.org/10.1016/j.jhazmat.2024.135422).

References

- [1] Ruiz-Jimenez, J., Raskala, S., Tanskanen, V., Aattela, E., Salkinoja-Salonen, M., Hartonen, K., et al., 2023. Evaluation of VOCs from fungal strains, building insulation materials and indoor air by solid phase microextraction arrow, thermal desorption-gas chromatography-mass spectrometry and machine learning approaches. *Environ Res* 224, 115494. <https://doi.org/10.1016/j.envres.2023.115494>.
- [2] Li, A.J., Pal, V.K., Kannan, K., 2021. A review of environmental occurrence, toxicity, biotransformation and biomonitoring of volatile organic compounds. *Environ Chem Ecotoxicol* 3, 91–116. <https://doi.org/10.1016/j.enceco.2021.01.001>.
- [3] Paciência, I., Madureira, J., Rufo, J., Moreira, A., Fernandes Ede, O., 2016. A systematic review of evidence and implications of spatial and seasonal variations of volatile organic compounds (VOC) in indoor human environments. *J Toxicol Environ Health B Crit Rev* 19 (2), 47–64. <https://doi.org/10.1080/10937404.2015.1134371>.
- [4] He, C., Cheng, J., Zhang, X., Douthwaite, M., Pattison, S., Hao, Z., 2019. Recent Advances in the Catalytic Oxidation of Volatile Organic Compounds: A Review Based on Pollutant Sorts and Sources. *Chem Rev* 119 (7), 4471–4568. <https://doi.org/10.1021/acs.chemrev.8b00408>.
- [5] Kang, S., Lee, K., Lim, M., 2022. Estimation of multi-route exposures to various chemicals during Children's clay toy use. *Environ Res* 212 (Pt D), 113500. <https://doi.org/10.1016/j.envres.2022.113500>.
- [6] Abplanalp, W., DeJarnett, N., Riggs, D.W., Conklin, D.J., McCracken, J.P., Srivastava, S., et al., 2017. Benzene exposure is associated with cardiovascular disease risk. *PLoS One* 12 (9), e0183602. <https://doi.org/10.1371/journal.pone.0183602>.
- [7] Maesano, C.N., Caillaud, D., Youssouf, H., Banerjee, S., Prud'Homme, J., Audi, C., et al., 2019. Indoor exposure to particulate matter and volatile organic compounds in dwellings and workplaces and respiratory health in French farmers. *Multidiscip Respir Med* 14, 33. <https://doi.org/10.1186/s40248-019-0194-3>.
- [8] Xiong, Y., Zhou, J., Xing, Z., Du, K., 2021. Cancer risk assessment for exposure to hazardous volatile organic compounds in Calgary, Canada. *Chemosphere* 272, 129650. <https://doi.org/10.1016/j.chemosphere.2021.129650>.
- [9] McGraw, K.E., Konkole, S.L., Riggs, D.W., Rai, S.N., DeJarnett, N., Xie, Z., et al., 2023. Exposure to Volatile Organic Compounds Is Associated with Hypertension in Black Adults: The Jackson Heart Study. *Environ Res* 223, 115384. <https://doi.org/10.1016/j.envres.2023.115384>.
- [10] Abouee-Mehrzi, A., Rasoulzadeh, Y., Solali, S., Esmaeilzadeh, H., Iraj, Z., Mesgari-Abbasi, M., 2023. Hematotoxicity induced by simultaneous exposure to noise and toluene in New Zealand white rabbits: Synergistic and antagonistic effects. *Toxicol Ind Health* 39 (8), 451–463. <https://doi.org/10.1177/07482337231180404>.
- [11] Liu, D., Zhao, N., Wang, M., Pi, X., Feng, Y., Wang, Y., et al., 2018. Urine volatile organic compounds as biomarkers for minimal change type nephrotic syndrome. *Biochem Biophys Res Commun* 496 (1), 58–63. <https://doi.org/10.1016/j.bbrc.2017.12.164>.
- [12] Gao, M., Wells, J.C.K., Johnson, W., Li, L., 2022. Socio-economic disparities in child-to-adolescent growth trajectories in China: findings from the China Health and Nutrition Survey 1991–2015. *Lancet Reg Health West Pac* 21, 100399. <https://doi.org/10.1016/j.lanwpc.2022.100399>.
- [13] NCD-RisC, 2020. Height and body-mass index trajectories of school-aged children and adolescents from 1985 to 2019 in 200 countries and territories: a pooled analysis of 2181 population-based studies with 65 million participants. *Lancet* 396 (10261), 1511–1524. [https://doi.org/10.1016/s0140-6736\(20\)31859-6](https://doi.org/10.1016/s0140-6736(20)31859-6).
- [14] Hu, X.J., Li, Z.H., Xie, L.N., Fu, H., Zhu, Y., 2024. [Determination of 35 antibiotics and four β -receptor agonists in teenagers and analysis of exposure status]. *Se Pu* 42 (2), 185–193. <https://doi.org/10.3724/sp.j.1123.2023.10022>.
- [15] Bradfield, J.P., Kember, R.L., Ulrich, A., Balkhiyarova, Z., Alyass, A., Aris, I.M., et al., 2024. Trans-ancestral genome-wide association study of longitudinal pubertal height growth and shared heritability with adult health outcomes. *Genome Biol* 25 (1), 22. <https://doi.org/10.1186/s13059-023-03136-z>.
- [16] Schell, L.M., 1991. Effects of pollutants on human prenatal and postnatal growth: Noise, lead, polychlorobiphenyl compounds, and toxic wastes, 34 (S13), 157–188. <https://doi.org/10.1002/ajpa.1330340609>.
- [17] Woodruff, T.J., 2024. Health effects of fossil fuel-derived endocrine disruptors. *N Engl J Med* 390 (10), 922–933. <https://doi.org/10.1056/NEJMr2300476>.
- [18] Brumberg, H.L., Karr, C.J., 2021. Ambient air pollution: health hazards to children. *Pediatrics* 147 (6), e2021051484. <https://doi.org/10.1542/peds.2021-051484>.
- [19] Wang, Y.F., Zhang, X.R., Zou, Y.X., 2023. Serum perfluoroalkyl substances and growth and development in US adolescents: a nationally representative cross-sectional study. *Eur J Pediatr* 182 (10), 4673–4681. <https://doi.org/10.1007/s00431-023-05136-4>.
- [20] Tahir, E., Cordier, S., Courtemanche, Y., Forget-Dubois, N., Desrochers-Couture, M., Bélanger, R.E., et al., 2020. Effects of polychlorinated biphenyls exposure on physical growth from birth to childhood and adolescence: a prospective cohort study. *Environ Res* 189, 109924. <https://doi.org/10.1016/j.envres.2020.109924>.
- [21] Gutiérrez-Delgado, R.I., Barraza-Villarreal, A., Escamilla-Núñez, M.C., Hernández-Cadena, L., Cortez-Lugo, M., Sly, P., et al., 2020. Prenatal exposure to VOCs and NOx and lung function in preschoolers. *Pedia Pulmonol* 55 (8), 2142–2149. <https://doi.org/10.1002/ppul.24889>.
- [22] Bobrowska-Korzeniowska, M., Majak, P., Brzozowska, A., Polańska, K., Kaleta, D., Smejda, K., et al., 2022. Cluster analysis of exhaled volatile organic compounds

- (VOCs)-link between environmental exposure and asthma in preschool children. *Clin Exp Allergy* 52 (8), 985–989. <https://doi.org/10.1111/cea.14175>.
- [23] Bland, V.L., Bea, J.W., Blew, R.M., Roe, D.J., Lee, V.R., Funk, J.L., et al., 2021. Influence of changes in soft tissue composition on changes in bone strength in peripubertal girls: the STAR longitudinal study. *J Bone Min Res* 36 (1), 123–132. <https://doi.org/10.1002/jbmr.4168>.
- [24] Yilmaz, D., Ersoy, B., Bilgin, E., Gümüşer, G., Onur, E., Pinar, E.D., 2005. Bone mineral density in girls and boys at different pubertal stages: relation with gonadal steroids, bone formation markers, and growth parameters. *J Bone Min Metab* 23 (6), 476–482. <https://doi.org/10.1007/s00774-005-0631-6>.
- [25] Bierhals, I.O., Vaz, J.D.S., Matijasevich, A., Barros, A.J.D., Tovo-Rodrigues, L., Barros, F.C., et al., 2022. Linear growth and relative weight gain in childhood and bone mass in adolescence: findings from the Pelotas (Brazil) 2004 birth cohort. *Arch Osteoporos* 18 (1), 5. <https://doi.org/10.1007/s11657-022-01189-4>.
- [26] El Hage, R., Moussa, E., Hammoud, A., Dandachi, G., Jacob, C., 2010. Birth weight is an independent determinant of whole body bone mineral content and bone mineral density in a group of Lebanese adolescent girls. *J Bone Min Metab* 28 (3), 360–363. <https://doi.org/10.1007/s00774-010-0165-4>.
- [27] Qadir, A., Liang, S., Wu, Z., Chen, Z., Hu, L., Qian, A., 2020. Senile osteoporosis: the involvement of differentiation and senescence of bone marrow stromal cells. *Int J Mol Sci* 21 (1), 349. <https://doi.org/10.3390/ijms21010349>.
- [28] Li, L., Liu, X., Zhang, X., Zhang, Y., Li, Q., Geng, H., et al., 2024. Chloroform associated with bone mineral density and bone mineral content in adults: a population-based cross-sectional research. *PLoS One* 19 (3), e0290132. <https://doi.org/10.1371/journal.pone.0290132>.
- [29] Kular, J., Tickner, J., Chim, S.M., Xu, J., 2012. An overview of the regulation of bone remodelling at the cellular level. *Clin Biochem* 45 (12), 863–873. <https://doi.org/10.1016/j.clinbiochem.2012.03.021>.
- [30] Zhou, H.L., Su, G.H., Zhang, R.Y., Di, D.S., Wang, Q., 2022. Association of volatile organic compounds co-exposure with bone health indicators and potential mediators. *Chemosphere* 308 (Pt 1), 136208. <https://doi.org/10.1016/j.chemosphere.2022.136208>.
- [31] Braegger, C., Campoy, C., Colomb, V., Decsi, T., Domellof, M., Fewtrell, M., et al., 2013. Vitamin D in the healthy European paediatric population. *J Pediatr Gastroenterol Nutr* 56 (6), 692–701. <https://doi.org/10.1097/MPG.0b013e31828f3c05>.
- [32] Kuraoka, S., Oda, M., Mitsubuchi, H., Nakamura, K., Katoh, T., Japan, E., et al., 2022. Impaired height growth associated with vitamin D deficiency in young children from the Japan Environment and Children's Study. *Nutrients* 14 (16), 3325. <https://doi.org/10.3390/nu14163325>.
- [33] Koo, W., Walyat, N., 2013. Vitamin D and skeletal growth and development. *Curr Osteoporos Rep* 11 (3), 188–193. <https://doi.org/10.1007/s11914-013-0156-1>.
- [34] Stounbjerg, N.G., Thams, L., Hansen, M., Larnkjær, A., Clerico, J.W., Cashman, K. D., et al., 2021. Effects of vitamin D and high dairy protein intake on bone mineralization and linear growth in 6- to 8-year-old children: the D-pro randomized trial. *Am J Clin Nutr* 114 (6), 1971–1985. <https://doi.org/10.1093/ajcn/nqab286>.
- [35] Xu, B., Feng, Y., Gan, L., Zhang, Y., Jiang, W., Feng, J., et al., 2021. Vitamin D status in children with short stature: accurate determination of serum vitamin D components using high-performance liquid chromatography-tandem mass spectrometry. *Front Endocrinol (Lausanne)* 12, 707283. <https://doi.org/10.3389/fendo.2021.707283>.
- [36] Xiao, P., Cheng, H., Wang, L., Hou, D., Li, H., Zhao, X., et al., 2023. Relationships for vitamin D with childhood height growth velocity and low bone mineral density risk. *Front Nutr* 10, 1081896. <https://doi.org/10.3389/fnut.2023.1081896>.
- [37] Huey, S.L., Acharya, N., Silver, A., Shen, R., Yu, E.A., Peña-Rosas, J.P., et al., 2020. Effects of oral vitamin D supplementation on linear growth and other health outcomes among children under five years of age. *Cochrane Database Syst Rev* 12 (12), Cd012875. <https://doi.org/10.1002/14651858.CD012875.pub2>.
- [38] Trilok-Kumar, G., Kaur, M., Rehman, A.M., Arora, H., Rajput, M.M., Chugh, R., et al., 2015. Effects of vitamin D supplementation in infancy on growth, bone parameters, body composition and gross motor development at age 3–6 years: follow-up of a randomized controlled trial. *Int J Epidemiol* 44 (3), 894–905. <https://doi.org/10.1093/ije/dyv116>.
- [39] Chowdhury, R., Taneja, S., Kvestad, I., Hysing, M., Bhandari, N., Strand, T.A., 2020. Vitamin D status in early childhood is not associated with cognitive development and linear growth at 6–9 years of age in North Indian children: a cohort study. *Nutr J* 19 (1), 14. <https://doi.org/10.1186/s12937-020-00530-2>.
- [40] Zaccagni, L., Rinaldo, N., Mazzoni, G., Mandini, S., Masotti, S., Toselli, S., et al., 2023. Assessing the impact of COVID-19 prevention measures on adolescent growth in Italy. *Healthc (Basel)* 11 (14), 2101. <https://doi.org/10.3390/healthcare11142101>.
- [41] Billionnet, C., Gay, E., Kirchner, S., Leynaert, B., Annesi-Maesano, I., 2011. Quantitative assessments of indoor air pollution and respiratory health in a population-based sample of French dwellings. *Environ Res* 111 (3), 425–434. <https://doi.org/10.1016/j.envres.2011.02.008>.
- [42] Weng, X., Tan, Y., Fei, Q., Yao, H., Fu, Y., Wu, X., et al., 2022. Association between mixed exposure of phthalates and cognitive function among the U.S. elderly from NHANES 2011–2014: three statistical models. *Sci Total Environ* 828, 154362. <https://doi.org/10.1016/j.scitotenv.2022.154362>.
- [43] Alwis, K.U., Blount, B.C., Britt, A.S., Patel, D., Ashley, D.L., 2012. Simultaneous analysis of 28 urinary VOC metabolites using ultra high performance liquid chromatography coupled with electrospray ionization tandem mass spectrometry (UPLC-ESI/MSMS). *Anal Chim Acta* 750, 152–160. <https://doi.org/10.1016/j.aca.2012.04.009>.
- [44] Feng, X., Qiu, F., Zheng, L., Zhang, Y., Wang, Y., Wang, M., et al., 2024. Exposure to volatile organic compounds and mortality in US adults: A population-based prospective cohort study. *Sci Total Environ* 928, 172512. <https://doi.org/10.1016/j.scitotenv.2024.172512>.
- [45] Tang, L., Liu, M., Tian, J., 2024. Volatile organic compounds exposure associated with depression among U.S. adults: Results from NHANES 2011–2020. *Chemosphere* 349, 140690. <https://doi.org/10.1016/j.chemosphere.2023.140690>.
- [46] de Onis, M., Garza, C., Victora, C.G., Onyango, A.W., Frongillo, E.A., Martines, J., 2004. The WHO multicentre growth reference study: planning, study design, and methodology. *Food Nutr Bull* 25 (1 Suppl), S15–S26. <https://doi.org/10.1177/15648265040251s103>.
- [47] Baran, D.T., Faulkner, K.G., Genant, H.K., Miller, P.D., Pacifici, R., 1997. Diagnosis and management of osteoporosis: guidelines for the utilization of bone densitometry. *Calcif Tissue Int* 61 (6), 433–440. <https://doi.org/10.1007/s002239900362>.
- [48] Wu, M., Si, J., Liu, Y., Kang, L., Xu, B., 2023. Association between composite dietary antioxidant index and hypertension: insights from NHANES. *Clin Exp Hypertens* 45 (1), 2233712. <https://doi.org/10.1080/10641963.2023.2233712>.
- [49] Ji, Y., Wei, C.B., Gu, W., Hou, L.L., 2023. Relevance of vitamin D on NAFLD and liver fibrosis detected by vibration controlled transient elastography in US adults: a cross-sectional analysis of NHANES 2017–2018. *Ann Med* 55 (1), 2209335. <https://doi.org/10.1080/07853890.2023.2209335>.
- [50] Signes-Pastor, A.J., Desai, G., García-Villarino, M., Karagas, M.R., Kordas, K., 2021. Exposure to a mixture of metals and growth indicators in 6–11-year-old children from the 2013–16 NHANES. *Expo Health* 13 (2), 173–184. <https://doi.org/10.1007/s12403-020-00371-8>.
- [51] Korovljev, D., Stajer, V., Ostojic, S.M., 2021. Relationship between dietary creatine and growth indicators in children and adolescents aged 2–19 years: a cross-sectional study. *Nutrients* 13 (3), 1027. <https://doi.org/10.3390/nu13031027>.
- [52] Zhao, L., Ogden, C.L., Yang, Q., Jackson, S.L., Loria, C.M., Galuska, D.A., et al., 2021. Association of usual sodium intake with obesity among US children and adolescents, NHANES 2009–2016. *Obes (Silver Spring)* 29 (3), 587–594. <https://doi.org/10.1002/oby.23102>.
- [53] Yu, L., Liu, W., Wang, X., Ye, Z., Tan, Q., Qiu, W., et al., 2022. A review of practical statistical methods used in epidemiological studies to estimate the health effects of multi-pollutant mixture. *Environ Pollut* 306, 119356. <https://doi.org/10.1016/j.envpol.2022.119356>.
- [54] Chen, Y., Wu, Y., Lv, J., Zhou, S., Lin, S., Huang, S., et al., 2024. Overall and individual associations between per- and polyfluoroalkyl substances and liver function indices and the metabolic mechanism. *Environ Int* 183, 108405. <https://doi.org/10.1016/j.envint.2023.108405>.
- [55] Lewiecki, E.M., Gordon, C.M., Bain, S., Leonard, M.B., Bishop, N.J., Bianchi, M.L., et al., 2008. International society for clinical densitometry 2007 adult and pediatric official positions. *Bone* 43 (6), 1115–1121. <https://doi.org/10.1016/j.bone.2008.08.106>.
- [56] Keil, A.P., Buckley, J.P., O'Brien, K.M., Ferguson, K.K., Zhao, S., White, A.J., 2020. A quantile-based g-computation approach to addressing the effects of exposure mixtures. *Environ Health Perspect* 128 (4), 47004. <https://doi.org/10.1289/ehp5838>.
- [57] Wu, M., Liu, M., Zhang, Y., Wu, J., Gao, M., Huang, F., et al., 2024. Serum HDL partially mediates the association between exposure to volatile organic compounds and kidney stones: a nationally representative cross-sectional study from NHANES. *Sci Total Environ* 907, 167915. <https://doi.org/10.1016/j.scitotenv.2023.167915>.
- [58] Klis, K., Wronka, I., 2020. Associations between childhood and adolescence exposure to air pollution and adult height in Polish women. *Environ Res* 189, 109965. <https://doi.org/10.1016/j.envres.2020.109965>.
- [59] Jedrychowski, W., Maugeri, U., Jedrychowska-Bianchi, I., 2002. Body growth rate in preadolescent children and outdoor air quality. *Environ Res* 90 (1), 12–20. <https://doi.org/10.1006/enrs.2002.4383>.
- [60] Lyu, X., Guo, H., Wang, Y., Zhang, F., Nie, K., Dang, J., et al., 2020. Hazardous volatile organic compounds in ambient air of China. *Chemosphere* 246, 125731. <https://doi.org/10.1016/j.chemosphere.2019.125731>.
- [61] Klepeis, N.E., Nelson, W.C., Ott, W.R., Robinson, J.P., Tsang, A.M., Switzer, P., et al., 2001. The National Human Activity Pattern Survey (NHAPS): a resource for assessing exposure to environmental pollutants. *J Expo Anal Environ Epidemiol* 11 (3), 231–252. <https://doi.org/10.1038/sj.jea.7500165>.
- [62] Forand, S.P., Lewis-Michl, E.L., Gomez, M.I., 2012. Adverse birth outcomes and maternal exposure to trichloroethylene and tetrachloroethylene through soil vapor intrusion in New York State. *Environ Health Perspect* 120 (4), 616–621. <https://doi.org/10.1289/ehp.1103884>.
- [63] Chang, M., Park, H., Ha, M., Hong, Y.C., Lim, Y.H., Kim, Y., et al., 2017. The effect of prenatal TVOC exposure on birth and infantile weight: the Mothers and Children's Environmental Health study. *Pedia Res* 82 (3), 423–428. <https://doi.org/10.1038/pr.2017.55>.
- [64] Katakura, Y., Kishi, R., Ikeda, T., Miyake, H., 2001. Effects of prenatal styrene exposure on postnatal development and brain serotonin and catecholamine levels in rats. *Environ Res* 85 (1), 41–47. <https://doi.org/10.1006/enrs.2000.4053>.
- [65] Edwards, M., Dai, R., Ahmed, S.A., 2018. Our Environment Shapes Us: The Importance of Environment and Sex Differences in Regulation of Autoantibody Production. *Front Immunol* 9, 478. <https://doi.org/10.3389/fimmu.2018.00478>.
- [66] Casimir, G.J., Heldenbergh, F., Hanssens, L., Mulier, S., Heinrichs, C., Lefevre, N., et al., 2010. Gender differences and inflammation: an in vitro model of blood cells stimulation in prepubescent children. *J Inflamm (Lond)* 7, 28. <https://doi.org/10.1186/1476-9255-7-28>.

- [67] Tanner, J.M., Whitehouse, R.H., Marubini, E., Resele, L.F., 1976. The adolescent growth spurt of boys and girls of the Harpenden growth study. *Ann Hum Biol* 3 (2), 109–126. <https://doi.org/10.1080/03014467600001231>.
- [68] Wang, Y., Yu, Y., Zhang, X., Zhang, H., Zhang, Y., Wang, S., et al., 2024. Combined association of urinary volatile organic compounds with chronic bronchitis and emphysema among adults in NHANES 2011–2014: The mediating role of inflammation. *Chemosphere* 361, 141485. <https://doi.org/10.1016/j.chemosphere.2024.141485>.
- [69] Melnick, R.L., Shackelford, C.C., Huff, J., 1993. Carcinogenicity of 1,3-butadiene. *Environ Health Perspect* 100, 227–236. <https://doi.org/10.1289/ehp.93100227>.
- [70] Zanlorenzi, S., Martins, P.C., Alves Junior, C.A.S., Castro, J.A.C., Lima, L.R.A., Petroski, E.L., et al., 2022. Association between bone mineral density and content and physical growth parameters among children and adolescents diagnosed with HIV: a cross-sectional study. *Sao Paulo Med J* 140 (5), 682–690. <https://doi.org/10.1590/1516-3180.2021.0549.R1.03012022>.
- [71] Zhou, S., Qiu, X.S., Zhu, Z.Z., Wu, W.F., Liu, Z., Qiu, Y., 2012. A single-nucleotide polymorphism rs708567 in the IL-17RC gene is associated with a susceptibility to and the curve severity of adolescent idiopathic scoliosis in a Chinese Han population: a case-control study. *BMC Musculoskelet Disord* 13, 181. <https://doi.org/10.1186/1471-2474-13-181>.
- [72] Agarwal, S., Misra, R., Aggarwal, A., 2008. Interleukin 17 levels are increased in juvenile idiopathic arthritis synovial fluid and induce synovial fibroblasts to produce proinflammatory cytokines and matrix metalloproteinases. *J Rheuma* 35 (3), 515–519.
- [73] Darmochwal-Kolarz, D., Chara, A., 2022. The Association of IL-17 and PlGF/sENG Ratio in Pre-Eclampsia and Adverse Pregnancy Outcomes. *Int J Environ Res Public Health* 20 (1), 768. <https://doi.org/10.3390/ijerph20010768>.
- [74] AlShwaimi, E., Berggreen, E., Furusho, H., Rossall, J.C., Dobeck, J., Yoganathan, S., et al., 2013. IL-17 receptor A signaling is protective in infection-stimulated periapical bone destruction. *J Immunol* 191 (4), 1785–1791. <https://doi.org/10.4049/jimmunol.1202194>.
- [75] Huang, Z., Pei, X., Graves, D.T., 2020. The Interrelationship Between Diabetes, IL-17 and Bone Loss. *Curr Osteoporos Rep* 18 (1), 23–31. <https://doi.org/10.1007/s11914-020-00559-6>.
- [76] Schett, G., 2011. Effects of inflammatory and anti-inflammatory cytokines on the bone. *Eur J Clin Invest* 41 (12), 1361–1366. <https://doi.org/10.1111/j.1365-2362.2011.02545.x>.
- [77] Sanderson, I.R., 2014. Growth problems in children with IBD. *Nat Rev Gastroenterol Hepatol* 11 (10), 601–610. <https://doi.org/10.1038/nrgastro.2014.102>.
- [78] Snega Priya, P., Pratiksha Nandhini, P., Arockiaraj, J., 2023. A comprehensive review on environmental pollutants and osteoporosis: Insights into molecular pathways. *Environ Res* 237 (Pt 2), 117103. <https://doi.org/10.1016/j.envres.2023.117103>.
- [79] Fossati, S., Valvi, D., Martinez, D., Cirach, M., Estarlich, M., Fernández-Somoano, A., et al., 2020. Prenatal air pollution exposure and growth and cardio-metabolic risk in preschoolers. *Environ Int* 138, 105619. <https://doi.org/10.1016/j.envint.2020.105619>.
- [80] Guo, P., Chen, Y., Wu, H., Zeng, J., Zeng, Z., Li, W., et al., 2020. Ambient air pollution and markers of fetal growth: a retrospective population-based cohort study of 2.57 million term singleton births in China. *Environ Int* 135, 105410. <https://doi.org/10.1016/j.envint.2019.105410>.
- [81] Song, K., Kwon, A., Chae, H.W., Suh, J., Choi, H.S., Choi, Y., et al., 2021. Vitamin D status is associated with bone mineral density in adolescents: findings from the Korea National Health and Nutrition Examination Survey. *Nutr Res* 87, 13–21. <https://doi.org/10.1016/j.nutres.2020.12.011>.
- [82] Uday, S., Manaseki-Holland, S., Bowie, J., Mughal, M.Z., Crowe, F., Höglér, W., 2021. The effect of vitamin D supplementation and nutritional intake on skeletal maturity and bone health in socio-economically deprived children. *Eur J Nutr* 60 (6), 3343–3353. <https://doi.org/10.1007/s00394-021-02511-5>.
- [83] Sudfeld, C.R., Manji, K.P., Smith, E.R., Aboud, S., Kisenye, R., Fawzi, W.W., et al., 2017. Vitamin D deficiency is not associated with growth or the incidence of common morbidities among Tanzanian infants. *J Pediatr Gastroenterol Nutr* 65 (4), 467–474. <https://doi.org/10.1097/mpg.0000000000001658>.
- [84] Chowdhury, R., Taneja, S., Bhandari, N., Kvstad, I., Strand, T.A., Bhan, M.K., 2017. Vitamin-D status and neurodevelopment and growth in young north Indian children: a secondary data analysis. *Nutr J* 16 (1), 59. <https://doi.org/10.1186/s12937-017-0285-y>.
- [85] Altowijri, A., Alloubani, A., Abdulhafiz, I., Saleh, A., 2018. Impact of nutritional and environmental factors on vitamin D deficiency. *Asian Pac J Cancer Prev* 19 (9), 2569–2574. <https://doi.org/10.22034/apjcp.2018.19.9.2569>.